PubChem Cross-Validation for ModelSEED Compounds

Improved Correction Logic with Quality Gates

ModelSEED Database Cleanup

March 22, 2026

Dataset Overview

Metric	Count
Total compounds with structures	37,004
Total compounds with names	45,170
Compounds with both name + InChIKey	30,333
Unique SMILES entries	36,978
Unique InChIKey entries	30,443
Unique InChI entries	30,443

Cross-validation performed against PubChem using three lookup strategies:

1. ChEBI xref lookup (highest confidence)
2. KEGG xref lookup
3. Compound name lookup (lowest confidence)

Validation Results (First 1,000 Compounds)

Category	Count
MATCH	828
PROTONATION_DIFF	109
MISMATCH	35
STEREO_DIFF	26
NOT_FOUND	2
Total	**1,000**

Category definitions:

MATCH InChIKey identical

PROTONATION_DIFF Same connectivity and stereo, different protonation (3rd block of InChIKey)

STEREO_DIFF Same connectivity, different stereochemistry (2nd block)

MISMATCH Different connectivity (1st block)

NOT_FOUND Not in PubChem

61 compounds flagged for correction (35 MISMATCH + 26 STEREO_DIFF).

	Total	**Accepted**	**Rejected**
MISMATCH (formula verified)	35	19	16
STEREO_DIFF (stereo checked)	26	5	21
Total	**61**	**24**	**37**

- **21 STEREO_DIFF rejected**: PubChem had fewer defined stereocenters (would lose stereochemistry)
- **16 MISMATCH rejected**: PubChem formula did not match stored formula (likely wrong compound mapping)
- **24 accepted**: Validated corrections that improve the data

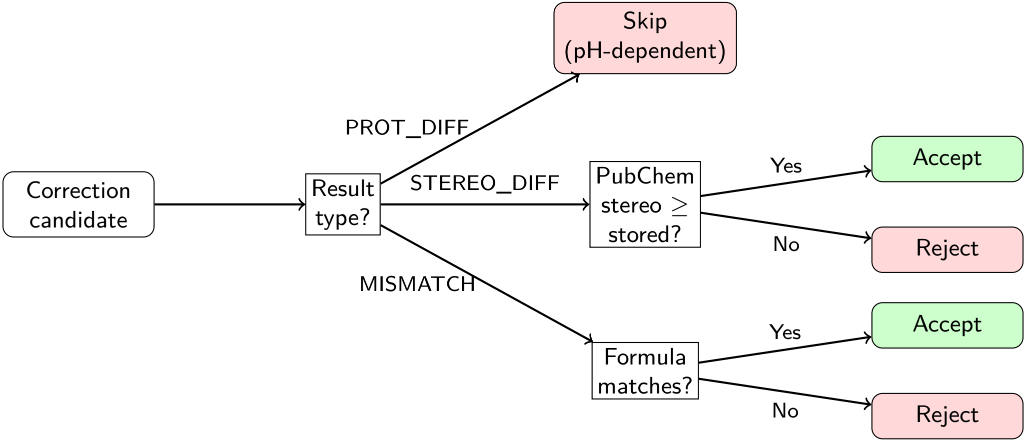
Corrections Applied (24 Compounds)

Field Updated	**Count**
SMILES	24
InChIKey	24
InChI	24
Formula/Charge	24*
Total field changes	**96**

- 19 MISMATCH corrections (formula verified against stored formula)
- 5 STEREO_DIFF corrections (PubChem added missing stereochemistry)

*Formula and Charge columns are recomputed from the corrected InChI to maintain internal consistency across all six columns.

Validation Gate: Decision Logic



Stereochemistry Gate: Examples

Corrections are only accepted when PubChem has $\geq$ defined stereocenters.

Compound	Name	Stored Stereo	PubChem Stereo	Decision
cpd00121	Inositol	6 / 6	0 / 6	Rejected
cpd00306	Xylitol	3 / 3	2 / 3	Rejected
cpd00366	Arabitol	3 / 3	2 / 3	Rejected
cpd00361	Acetoin	0 / 1	1 / 1	Accepted
cpd01049	Atropine	3 / 4	2 / 4	Rejected

Format: defined / potential stereocenters. PubChem often stores the parent compound without resolved stereochemistry, while ModelSEED stores the specific biological stereoisomer.

Formula Gate: Examples

Corrections are only accepted when the molecular formula matches.

Compound	**Name**	**Stored**	**PubChem**	**Decision**
cpd00151	Thymine	$C_5H_6N_2O_2$	$C_5H_6N_2O_2$	Accepted
cpd00204	Carbon monoxide	CHO	CO	Rejected
cpd00250	Creatine	$C_4H_9N_3O_2$	$C_4H_9N_3O_2$	Accepted
cpd00271	Ferricyanide	C_6FeN_6	C_6FeN_6	Accepted

cpd00204: ChEBI:17245 maps to neutral CO in PubChem (CID 281), but the stored structure is a protonated form (CHO^+). The formula mismatch reveals an incorrect xref mapping.

Cross-Validation: xref and Name Lookups

xref cross-validation: Compounds with both ChEBI and KEGG identifiers are queried against both databases. Conflicting CIDs are flagged as `XREF_CONFLICT` and excluded from auto-correction.

Multi-name cross-validation: Up to 3 compound names are queried. If different names return different PubChem CIDs, the compound is marked `AMBIGUOUS` and excluded.

New Status	Meaning
XREF_CONFLICT	ChEBI and KEGG xrefs point to different PubChem compounds
AMBIGUOUS	Multiple compound names resolve to different PubChem CIDs

Both categories require manual review before any correction is applied.

Data Preservation

- Timestamped backup before every write
- Corrections log appended, not overwritten
- ISO 8601 timestamps on every entry
- Backup auto-removed if no changes

Formula/Charge Consistency

- Recomputed from corrected InChI
- Uses BiochemPy InChI parser
- All rows for a compound updated

Cache Management

- SQLite cache for resumability
- `--clear-cache` for fresh runs
- Rate-limited PubChem API access

Output Reports

- `pubchem_validation_report.tsv`
- `pubchem_mismatches.tsv`
- `pubchem_protonation_diffs.tsv`
- `pubchem_corrections_log.tsv`

Summary of Improvements

Feature	Description
Stereo-aware gate	Count stereocenters via RDKit; reject if PubChem has fewer
Formula cross-check	Parse InChI formula; reject if different molecule
Formula/Charge sync	Recompute from corrected InChI; keep all columns consistent
ChEBI/KEGG cross-check	Query both sources; flag conflicts
Multi-name validation	Try 3 names; flag ambiguous mappings
Name dedup fix	Handle multiple compounds sharing the same name
Timestamped backup	Full copy before any modification
Append-mode log	Preserve audit history across runs
Protonation report	Separate file for pH-dependent differences
Cache clearing	`--clear-cache` for fresh re-validation